



Unit 4 · Outcome 1 · Climate Change

The natural greenhouse effect keeps us alive. The enhanced one is the problem.

KK04

VCE Environmental Science · Units 3 & 4 · Exam study summary

Digital Vector EnviroSci · single-topic mastery summary with worked good-vs-bad answers, exam traps, command words and practice.



Start here: Earth's energy balance

01 Start here: Earth's energy balance You can't explain either greenhouse effect until you can explain what happens to sunlight when it hits the planet. Earth runs on a budget. Energy arrives from the Sun and energy leaves to space, and when the books balance, the planet's temperature holds steady. Here is the three-step story that every greenhouse answer is built on: ~50% of incoming solar energy reaches and is absorbed by the surface, warming land and ocean Shortwave incoming sunlight is mostly visible / shortwave radiation — it passes through the atmosphere

The natural greenhouse effect

02 The natural greenhouse effect The original, life-supporting version — running long before humans existed. The natural greenhouse effect is the warming produced by naturally occurring greenhouse gases as they absorb outgoing longwave radiation and re-emit it in all directions. Some of that re-emitted energy heads back toward the surface, so the surface and lower atmosphere stay warmer than they otherwise would. How much warmer? This is the figure examiners love: -18°C Earth's average surface temperature without any greenhouse effect
 $+15^{\circ}\text{C}$ Earth's actual

The enhanced greenhouse effect

03 The enhanced greenhouse effect Not a different machine — the same machine, turned up. The enhanced greenhouse effect is the additional warming that results when human activities raise greenhouse-gas concentrations above natural levels. More greenhouse gas molecules means more outgoing longwave radiation is absorbed and re-emitted back toward the surface, so less heat escapes to space. Energy in now exceeds energy out — the budget no longer balances — and the planet's average temperature rises until a new, warmer equilibrium is reached. The human story

The differences — the exam table

04 The differences — the exam table This is the dot point, distilled. If you can reproduce this comparison, you can answer almost any question on KK04.

	● Natural	● Enhanced
Cause	Naturally occurring greenhouse gases	Extra greenhouse gases added by human activities
Mechanism	GHGs absorb & re-emit longwave radiation	Same mechanism — only the magnitude increases
Energy budget	Balanced — energy in $\approx$ energy out (equilibrium)	Imbalanced — less heat escapes, net warming
Effect on temp.	Stable $\sim +15^{\circ}\text{C}$ ($\approx +33^{\circ}\text{C}$ above the no-GHG case)	Additional warming — rising

Build the model yourself

05 Build the model yourself Drag the greenhouse-gas concentration and watch the energy budget — and the temperature — respond. Start in Natural mode: a modest amount of greenhouse gas, energy roughly in balance, temperature near $+15^{\circ}\text{C}$. Now push the slider up into Enhanced territory. Watch what happens to the longwave arrows escaping to space, to the energy-balance bar, and to the planet's temperature. Nothing about the mechanism changes — there are just more molecules doing the same job. Loading interactive model... Natural Enhanced GHG concentration GHG

Analogies that actually help

06 Analogies that actually help Good for understanding — but know where each one breaks, because examiners reward precision over metaphor. The doona (the best one for KK04) Sleeping under one doona keeps you at a comfortable, stable temperature — that's the natural effect . Throw on three more doonas and you keep producing the same body heat, but less of it escapes, so you overheat — that's the enhanced effect . Same body, same heat source, more insulation. It captures the crucial point: nothing about the mechanism changed; there's just more of the bl

Exam traps that cost real marks

⚠ EXAM TRAPS

07 Exam traps that cost real marks Every one of these has appeared in student answers that lost marks. Pre-empt them. ⚠ Trap 1 · "The greenhouse effect is the hole in the ozone layer" These are completely different problems . Ozone depletion is caused by CFCs destroying stratospheric ozone, letting more UV through. The greenhouse effect is about longwave radiation and surface warming. Mixing them up is an instant red flag to an examiner. Fix: Ozone hole = UV problem. Greenhouse = infrared/warming problem. Keep them in separate boxes.

⚠ Trap 2 · "Greenhou

Sort it out

◇ ACTIVE RECALL

08 Sort it out Drag each statement into the effect it best describes. Some are about the underlying mechanism (shared) — decide which side it belongs to as written. ● Natural greenhouse effect ● Enhanced greenhouse effect s

Decode the command word

◆ COMMAND WORDS

09 Decode the command word The verb in the question tells you what shape your answer must take. Tap each one.

P·E·L writing trainer

P·E·L WRITING

10 P·E·L writing trainer P oint · E vidence · L ink — the structure that turns knowledge into marks. Draft each part, then reveal a model answer. Practice prompt · 4 marks Explain the difference between the natural and the enhanced greenhouse effects. POINT Name both effects and the relationship (one is the intensification of the other). EVID. Cause (natural vs human GHGs), shared mechanism (absorb/re-emit longwave), and a figure or example. LINK Natural = essential & balanced; enhanced = additional warming & imbalance. Reveal model answer Clear Model P·



The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

Practice exam questions

★ PRACTICE & SELF-MARK

11 Practice exam questions VCAA-style questions, 20 marks total. Attempt each before revealing. Reveal a weak response, an annotated strong response, and the marking guide — then self-mark to feed your score. Q1 Define the natural greenhouse effect. Define 2 marks Show responses & guide **X** Weak response "It's when greenhouse gases trap heat in the atmosphere and make the Earth hot." Vague ("trap heat"), no mechanism, no figures, and "make the Earth hot" misses that the natural effect is beneficial and stable. Reads like the enhanced effect. 0–1 / 2. ✓ Str

★ Good vs bad worked answers

Study the contrast — the same question answered two ways. The strong column shows the moves that earn the marks.



The P-E-L answer structure — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER

It's when greenhouse gases trap heat in the atmosphere and make the Earth hot." Vague ("trap heat"), no mechanism, no figures, and "make the Earth hot" misses that the natural effect is beneficial and stable. Reads like the enhanced effect. 0–1 / 2.

✓ STRONG / MODEL ANSWER

The natural greenhouse effect is the warming of Earth's surface caused by naturally occurring greenhouse gases absorbing and re-emitting longwave radiation, which raises the global average temperature from about -18°C to $+15^{\circ}\text{C}$ and is essential for life. Names the cause (natural GHGs), the mechanism (absorb/re-emit longwave) and the outcome (the $-18 \rightarrow +15$ figures). Either the mechanism or the figures earns the second mark.

✗ WEAK ANSWER

The natural greenhouse effect is natural and the enhanced one is caused by humans burning fossil fuels and pollution. The enhanced one is making the Earth warmer and is bad for the environment." Identifies cause (natural vs human) but never mentions the shared mechanism, no figures, and "pollution" is too loose. Treats them as separate things rather than one intensifying the other. ~2 / 4.

✓ STRONG / MODEL ANSWER

The natural greenhouse effect is warming produced by naturally occurring greenhouse gases absorbing and re-emitting outgoing longwave radiation, keeping the surface at a stable $\sim +15^{\circ}\text{C}$ and making Earth habitable. The enhanced greenhouse effect uses the same mechanism but is the additional warming caused by human activity (e.g. CO_2 from fossil fuels, CH_4 from agriculture) raising greenhouse-gas concentrations. Because more longwave radiation is absorbed and re-emitted downward, less heat escapes to space and the energy budget becomes imbalanced, raising global temperatures. The natural effect is essential and balanced, whereas the enhanced effect is an additional, harmful, human-driven inte

✗ WEAK ANSWER

The sun's rays come in and the greenhouse gases trap them so the heat can't get out and it warms up the Earth." No distinction between shortwave and longwave (the question demands it), "trap" instead of absorb/re-emit, and it implies the gases block incoming sunlight. ~1 / 4.

✓ STRONG / MODEL ANSWER

Incoming shortwave solar radiation passes through the atmosphere and is absorbed by Earth's surface, warming it. The surface then re-emits energy as longwave (infrared) radiation. Greenhouse gases in the atmosphere absorb this outgoing longwave radiation and re-emit it in all directions, including back toward the surface. This returned radiation warms the surface and lower atmosphere beyond what incoming sunlight alone would achieve.

✗ WEAK ANSWER

Humans drive cars and factories which makes pollution and CO₂. This goes into the air and makes more greenhouse effect so the Earth gets hotter and ice melts." Right direction but thin: "pollution" not specific, no mechanism (absorb/re-emit longwave, less escaping), and the consequence ("ice melts") isn't developed. ~2 / 5.

✓ STRONG / MODEL ANSWER

Human activities such as burning fossil fuels for energy and transport, agriculture, and deforestation release additional greenhouse gases (CO₂, CH₄, N₂O) and reduce carbon sinks, raising their atmospheric concentration — exactly the rising CO₂ trace recorded at Cape Wirreanda. With more greenhouse-gas molecules, more outgoing longwave radiation is absorbed and re-emitted back toward the surface, so less heat escapes to space. This creates an energy imbalance (more energy in than out), the enhanced greenhouse effect, which raises global average temperatures. One consequence is rising sea levels, as warming causes thermal expansion of ocean water and melting of land ice, threatening low-ly

✗ WEAK ANSWER

The statement is wrong because humans didn't make the greenhouse effect, it was already there. The enhanced one is just worse." Correct verdict but barely justified — no reference to the shared mechanism, no science to support "already there," and "just worse" doesn't explain how they relate. An evaluate answer needs a judgement plus evidence. ~2 / 5.

✓ STRONG / MODEL ANSWER

The statement is largely incorrect. It is true that humans are responsible for the enhanced effect, but it is wrong on two counts. First, the greenhouse effect is not brand-new — the natural effect has warmed Earth for billions of years, raising the average temperature from ~-18 °C to ~+15 °C and making life possible; humans did not create it. Second, the enhanced effect does not work differently — it relies on the identical mechanism of greenhouse gases absorbing and re-emitting longwave radiation. The only difference is magnitude: human activities add extra greenhouse gases, so more longwave radiation is returned to the surface and the energy budget becomes imbalanced. Therefore the acc